



Particle Globe System — Reusable Implementation Guide



Overview

This document defines a **high-performance, interactive 3D particle globe** built with Three.js.

Core Features:

- Fibonacci sphere particle distribution
- Cursor-based magnetic interaction
- Scroll-driven animation system
- Cinematic bloom (post-processing)
- Depth via fog and camera motion
- Smooth physics-based particle behavior



Core Configuration

```
PARTICLE_COUNT = 4000  
RADIUS = 1.6
```

Performance Notes:

- 3000–5000 particles = optimal range
- Pixel ratio capped at **1.5**
- Uses GPU-friendly `PointsMaterial`



Geometry Construction (Fibonacci Sphere)

Particles are evenly distributed using a **golden angle spiral**.

Key Formula:

```
phi = Math.PI * (3 - Math.sqrt(5))
```

Behavior:

- Uniform distribution across sphere
- No clustering (unlike random placement)

Rendering Setup

Renderer:

- Antialias enabled
- SRGB color space
- Dark background (#0a0a0a)

Scene:

- Fog enabled for depth illusion

```
scene.fog = new THREE.Fog(0x0a0a0a, 3, 8)
```

Material System

PointsMaterial Configuration:

- Size:
- Additive blending
- Vertex colors enabled
- Transparent with soft opacity

Visual Effect:

- Glow-like energy particles
- Smooth blending (no harsh edges)

Color Strategy

Each particle gets a slight variation:

```
R: 0  
G: 0.6   
B: 1
```

Result:

- Blue → cyan gradient
 - Organic, non-uniform appearance
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Post-Processing (Cinematic Glow)

Unreal Bloom Pass:

- Strength: 0.9
- Radius: 0.5
- Threshold: 0.1

Effect:

- Soft glow around particles
 - Premium “energy field” look
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Cursor Interaction System

Behavior:

Particles respond to cursor with:

1. Attraction (Magnet Effect)

- Pull toward cursor
- Strength based on distance

```
force = ((maxDist - dist) / maxDist)^2
```

2. Repulsion (Close Range)

- Prevents collapse into cursor
- Adds “bubble” behavior

3. Depth Reaction

- Particles move along Z-axis
 - Creates true 3D interaction
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Ripple System

Particles generate wave motion:

```
ripple = sin(dist * 10 - time) * 0.02
```

Effect:

- Cursor creates energy waves
 - Feels fluid and organic
-

Scroll Interaction System

Scroll Progress:

```
scrollProgress = clamp(0  1)
```

Based on:

- Hero section visibility
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Effects Controlled by Scroll:

1. Camera Zoom

```
camera.z = 5  3.5
```

2. Particle Spread

```
spread = 1  1.3
```

3. Magnet Strength

- Increased with scroll

4. Pulse Intensity

- Stronger “breathing” effect
-

Camera System

Features:

1. Mouse Parallax

- Smooth X/Y drift
- Adds cinematic motion

2. Scroll Zoom

- Moves closer on scroll
 - Increases immersion
-

Animation System

Continuous Effects:

1. Rotation

- Slow Y-axis spin

2. Breathing Pulse

```
scale = 1 + sin(time) * intensity
```

3. Particle Physics Loop

- Runs every frame
 - Updates all particle positions
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Particle Physics Model

Each frame:

1. Calculate distance to cursor
 2. Apply magnetic force
 3. Apply repulsion (if close)
 4. Apply ripple effect
 5. Interpolate back to original position
 6. Apply scroll-based spread
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Performance Optimization

Critical Rules:

- Cap pixel ratio:

```
Math.min(devicePixelRatio, 1.5)
```

- Limit particle count:

MAX: ~5000

- Use BufferGeometry (NOT standard geometry)
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Cleanup (React Lifecycle)

On unmount:

- Cancel animation frame
 - Remove all event listeners
 - Dispose:
 - Renderer
 - Composer
 - Remove canvas from DOM
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Final Experience

User interaction flow:

1. Page loads → calm floating globe
 2. Mouse moves → particles react like magnetic fluid
 3. Scroll begins → system “activates”
 4. Scroll deeper → increased energy, motion, depth
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Extension Ideas (Optional)

1. Neural Connections

- Lines between nearby particles

2. Audio Reactivity

- Motion based on sound input

3. Scroll Storytelling

- Different states per section

4. Shader Upgrade

- Custom GLSL for advanced visuals
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Conceptual Summary

This system simulates:

- Magnetic field
- Liquid surface
- Neural network
- Energy sphere

👉 Designed to represent: **AI + Intelligence + Connectivity**

Usage

Use this system when building:

- AI company websites
 - Creative agency portfolios
 - Landing pages needing “wow factor”
 - Interactive hero sections
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Key Principle

Motion should communicate meaning — not just decoration.
